ACCRA TECHNICAL UNIVERSITY

FACULTY OF APPLIED SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

BCS 404 – INTRODUCTION TO DATA SCIENCE WITH PYTHON

EXPLORATORY DATA ANALYSIS, STATISTICAL ANALYSIS AND MACHINE LEARNING USING THE TITANIC DATASET

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CHAPTER ONE

1. Introduction

1.1 Background

1. Data science is the process of collecting, cleaning, analyzing, and interpreting data to discover meaningful information and support decision-making. It combines programming, statistics, and machine learning techniques to identify patterns and make predictions from data. With the increasing amount of data generated every day, data science has become an important tool in solving real-world problems across different industries, including healthcare, finance, education, and transportation.

2. The Titanic dataset is one of the most widely used datasets for learning data science and machine learning. It contains information about passengers aboard the RMS Titanic, such as age, gender, passenger class, fare paid, and survival status. The dataset is commonly used because it provides a practical example of how data can be explored, cleaned, analyzed, visualized, and used to build predictive models.

3. In this project, the Titanic dataset obtained from Kaggle was analyzed using Python. Several Python libraries, including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn, were used to perform data preprocessing, visualization, statistical analysis, and machine learning. A Logistic Regression model was developed to predict whether a passenger survived the Titanic disaster based on selected features

1.2 Objectives

The objectives of this project are to:

- Import and explore the Titanic dataset using Python.
- Clean and preprocess the dataset by handling missing values and duplicates.
- Perform exploratory data analysis using charts and graphs.
- Conduct statistical analysis to identify relationships among variables.
- Develop a Logistic Regression model to predict passenger survival.
- Evaluate the performance of the machine learning model using suitable evaluation metrics.

CHAPTER TWO

2.0 Dataset Description

The dataset used for this project is the **Titanic dataset**, which is publicly available on Kaggle. It contains information about passengers who travelled on the RMS Titanic during its maiden voyage in 1912. The dataset is widely used in data science and machine learning because it provides real-world data suitable for classification and predictive analysis.

The dataset contains information such as passenger class, age, gender, fare paid, family relationships, port of embarkation, and whether the passenger survived the disaster. These variables make it possible to analyze the factors that influenced survival and to build a machine learning model that predicts passenger survival.

2.1 Source of the Dataset

The dataset was obtained from the Kaggle Titanic Machine Learning Competition.

Source:
https://www.kaggle.com/competitions/titanic/data

The dataset is intended to help learners practice data cleaning, visualization, statistical analysis, and machine learning techniques using real-world data.

2.2 Dataset Size

The Titanic training dataset contains:

- **891 observations (rows) representing passengers.**
- **12 variables (columns) describing passenger information.**

2.3 Description of the Variables

Variable	**Description**
PassengerId	A unique identification number assigned to each passenger.
Survived	Indicates whether the passenger survived (1) or did not survive (0). This is the target variable.

Variable	Description
Pclass	Passenger class (1 = First Class, 2 = Second Class, 3 = Third Class).
Name	Full name of the passenger.
Sex	Gender of the passenger (Male or Female).
Age	Age of the passenger in years.
SibSp	Number of siblings or spouses travelling with the passenger.
Parch	Number of parents or children travelling with the passenger.
Ticket	Passenger ticket number.
Fare	Amount paid for the ticket.
Cabin	Cabin number assigned to the passenger.
Embarked	Port where the passenger boarded the ship (C = Cherbourg, Q = Queenstown, S = Southampton).

2.4 Why the Titanic Dataset Was Selected

The Titanic dataset was selected because it is one of the most widely used datasets for learning data science and machine learning. It contains both numerical and categorical variables, making it suitable for data preprocessing, visualization, statistical analysis, and classification tasks.

The dataset also presents a real-world problem of predicting passenger survival, allowing different analytical techniques to be applied in order to identify the factors that influenced survival during the Titanic disaster.

2.5 Target Variable

The target variable used in this project is **Survived**.

- **0 = Passenger did not survive.**
- **1 = Passenger survived.** The machine learning model was developed to predict this variable using the remaining relevant passenger information as predictor variables

CHAPTER THREE

3.0 Methodology

This chapter explains the procedures and techniques used to complete the project. The methodology includes data acquisition, data cleaning, data visualization, statistical analysis, and machine learning. Python and its data science libraries were used to perform all the analyses.

3.1 Data Acquisition

The first step of the project involved obtaining the Titanic dataset from Kaggle. The dataset was downloaded and imported into Jupyter Notebook using the Pandas library. After importing the dataset, an initial exploration was carried out to understand its structure and contents.

The following activities were performed:

- Imported the Titanic dataset into Python.
- Displayed the number of rows and columns (dataset dimensions).
- Displayed the names of all variables (column names).
- Displayed the first five records of the dataset.
- Checked the data type of each variable.

This initial exploration helped in understanding the dataset before any preprocessing or analysis was performed.

3.2 Data Cleaning

Data cleaning was performed to improve the quality of the dataset before analysis and model development. Incomplete and inconsistent data can negatively affect statistical analysis and machine learning performance.

The following preprocessing steps were carried out:

Detection of Missing Values

The dataset was checked for missing values in each column using Pandas functions. Missing values were found in variables such as **Age**, **Cabin**, and **Embarked**.

Handling Missing Values

Appropriate methods were used to handle the missing values.

- Missing values in **Age** were replaced with the median age because age is a numerical variable and the median is less affected by extreme values.
- Missing values in **Embarked** were replaced with the most frequently occurring value (mode).
- The **Cabin** column contained a large number of missing values. Depending on the preprocessing approach used in the notebook, the column was either removed or the missing values were handled appropriately because the missing values were too many to provide reliable information.

Detection of Duplicate Records

The dataset was examined for duplicate observations using Pandas.

Removal of Duplicates

Duplicate records were removed where necessary to avoid repeated observations that could affect the analysis and model performance.

After completing these preprocessing steps, the dataset became clean and suitable for visualization, statistical analysis, and machine learning.

3.3 Data Visualization

Exploratory Data Analysis (EDA) was performed using visualization techniques to better understand the characteristics of the dataset and identify relationships among variables.

The following visualizations were produced:

Histogram of Passenger Ages

A histogram was created to show the distribution of passenger ages. This helped identify the age groups with the highest number of passengers and determine whether the ages were normally distributed.

Bar Chart of Passenger Class

A bar chart was used to display the number of passengers in each passenger class (First, Second, and Third Class). This visualization made it easy to compare the class distribution.

Boxplot of Age by Passenger Class

A boxplot was produced to compare the age distribution across different passenger classes and identify the presence of outliers.

Scatter Plot of Age versus Fare

A scatter plot was created to examine the relationship between passenger age and ticket fare. This helped determine whether any relationship existed between the two variables.

Correlation Heatmap

A heatmap was generated to visualize the correlation coefficients among the numerical variables. Strong positive and negative relationships were easily identified using different color intensities.

Pairplot

A pairplot was produced to examine relationships among selected numerical variables simultaneously. It also helped identify possible patterns, clusters, and trends within the dataset.

3.4 Statistical Analysis

Statistical analysis was performed to summarize the dataset and identify important relationships among variables.

The analyses included:

Descriptive Statistics

Measures such as the mean, median, standard deviation, minimum value, maximum value, and quartiles were computed to summarize the numerical variables.

Frequency Distribution

Frequency tables were generated for categorical variables such as passenger class, gender, and survival status. These tables showed how observations were distributed across different categories.

Correlation Analysis

Correlation analysis was performed to determine the strength and direction of relationships between numerical variables. Correlation coefficients ranged from -1 to +1, where values closer to +1 indicate a strong positive relationship, values closer to -1 indicate a strong negative relationship, and values close to zero indicate little or no relationship.

3.5 Machine Learning

A Logistic Regression classification model was developed to predict passenger survival.

The following steps were followed:

Selection of Predictor Variables

Relevant variables that influence passenger survival were selected as input features. The target variable was **Survived**.

Data Splitting

The dataset was divided into two parts:

- **Training set (80%)** for training the model.
- **Testing set (20%)** for evaluating the model.

Model Training

The Logistic Regression algorithm from the Scikit-learn library was trained using the training dataset.

Model Prediction

The trained model was used to predict the survival status of passengers in the testing dataset.

Model Evaluation

The performance of the model was evaluated using:

- Accuracy Score
- Confusion Matrix
- Classification Report

These evaluation metrics were used to determine how well the Logistic Regression model predicted passenger survival.

CHAPTER FOUR

4.0 Results

This chapter presents the results obtained from the data acquisition, data cleaning, data visualization, statistical analysis, and machine learning processes performed on the Titanic dataset.

4.1 Task 1: Data Acquisition

The Titanic dataset was successfully downloaded from Kaggle and imported into Python using the Pandas library. Initial exploration was performed to understand the structure and contents of the dataset.

Dataset Dimensions

The dataset contains **891 rows** and **12 columns**.

Interpretation

This indicates that the dataset contains information on 891 passengers described by 12 different variables.

Column Names

The dataset contains the following variables:

- PassengerId
- Survived
- Pclass
- Name
- Sex
- Age
- SibSp
- Parch
- Ticket
- Fare
- Cabin
- Embarked

Interpretation

These variables describe the passengers' demographic information, travel details, and survival status

First Five Observations

The first five rows of the dataset were displayed using the head() function.

Interpretation

Displaying the first five records helped verify that the dataset had been imported successfully and provided an overview of the available information.

Insert Screenshot: *First Five Rows (df.head()).*

Data Types

The data types of all variables were displayed using the info() function.

The dataset contains:

- Integer variables
- Floating-point variables
- Object (text) variables

Interpretation

Understanding the data types helped determine which variables required preprocessing before analysis and machine learning.

Insert Screenshot: *Output of df.info().*

4.2 Task 2: Data Cleaning

The dataset was cleaned to improve its quality before analysis.

Detection of Missing Values

The dataset was checked for missing values.

Missing values were found mainly in:

- Age
- Cabin
- Embarked

Interpretation

Missing values can reduce the accuracy of statistical analysis and machine learning models if they are not handled properly.

Insert Screenshot: *Output of df.isnull().sum().*

Handling Missing Values

The missing values were handled as follows:

- Missing values in **Age** were replaced with the median value.
- Missing values in **Embarked** were replaced with the most frequent value (mode).
- The **Cabin** column was removed (or handled appropriately) because it contained a large number of missing values.

Interpretation

These preprocessing steps improved the quality of the dataset and ensured that the machine learning model could be trained without errors.

Detection of Duplicate Records

The dataset was checked for duplicate records.

Result

No duplicate records were found.

(If your notebook found duplicates, replace this statement with your actual result.)

Removal of Duplicate Records

Since no duplicate records were present, no rows were removed from the dataset.

Interpretation

This indicates that every observation represented a unique passenger.

Summary of Data Cleaning

The preprocessing stage successfully prepared the dataset for further analysis by:

- Handling missing values.

- Checking duplicate records.
- Ensuring the dataset was clean and consistent.

CHAPTER FIVE

DISCUSSION

This chapter discusses the findings obtained from the exploratory data analysis, statistical analysis, and machine learning model developed using the Titanic dataset.

5.1 Discussion of Exploratory Data Analysis

The exploratory data analysis provided useful insights into the characteristics of the Titanic dataset. Data cleaning was carried out by handling missing values and removing any duplicate records to improve the quality of the dataset before analysis.

The visualizations revealed several important patterns. The histogram of passenger ages showed that most passengers were young adults, although passengers of different age groups were represented. The passenger class distribution indicated that Third Class (Pclass 3) had the highest number of passengers, while First Class had the fewest.

The boxplot comparing passenger age across different classes showed variations in age distribution among the classes. The scatter plot of Age versus Fare indicated that ticket fares varied greatly, with some passengers paying significantly higher fares than others regardless of age.

The correlation heatmap and pairplot further helped identify relationships between numerical variables and provided a better understanding of the dataset before building the prediction model.

5.2 Discussion of Statistical Analysis

The descriptive statistics summarized the central tendency and variation of important variables such as Age and Fare. Frequency distributions showed that more passengers travelled in Third Class than in First or Second Class.

Correlation analysis identified relationships among the numerical variables. The strongest positive correlation was observed between Fare and Passenger Class-related variables (depending on your results), while negative correlations suggested inverse relationships between some variables.

Overall, the statistical analysis confirmed that passenger characteristics such as ticket class and fare appeared to influence survival outcomes.

5.3 Discussion of Machine Learning Results

A Logistic Regression classifier was developed to predict passenger survival using selected predictor variables. The dataset was divided into training and testing sets before model training.

The model achieved a good prediction accuracy, indicating that Logistic Regression can effectively classify passenger survival using the available features. The confusion matrix showed that the model correctly classified most passengers, although a small number of incorrect predictions were made.

The classification report demonstrated satisfactory values for precision, recall, and F1-score, suggesting that the model performed reasonably well in distinguishing between survivors and non-survivors.

Overall, the machine learning model successfully demonstrated how classification techniques can be applied to real-world datasets for predictive analysis.

5.4 Limitations of the Study

The study had several limitations:

- The Titanic dataset contains missing values that required preprocessing before analysis.
- Only a Logistic Regression model was used; other machine learning algorithms may produce higher prediction accuracy.
- Some potentially useful variables required additional feature engineering which was beyond the scope of this project.
- The model was trained using historical data from a single event, limiting its generalization to other datasets.

5.5 Recommendations

Based on the findings of this study, the following recommendations are made:

- Future studies should compare multiple machine learning algorithms such as Decision Trees, Random Forest, and Support Vector Machines.
- Additional feature engineering should be performed to improve prediction performance.
- Larger and more diverse datasets should be considered to improve model generalization.

- More advanced evaluation techniques such as cross-validation can be used to obtain more reliable performance estimates.

CHAPTER SIX

CONCLUSION

This project successfully applied data science techniques to the Titanic dataset using Python. The project involved data acquisition, data cleaning, exploratory data analysis, statistical analysis, and the development of a machine learning classification model.

The exploratory analysis provided valuable insights into passenger demographics and travel characteristics, while the statistical analysis identified important relationships among the variables. The Logistic Regression model successfully predicted passenger survival with good performance, demonstrating the usefulness of supervised machine learning for classification problems.

Overall, the project achieved its objectives and demonstrated practical skills in data preprocessing, visualization, statistical analysis, and predictive modelling using Python.

References

Use APA 7th edition. Here are suitable references:

1. Géron, A. (2022). *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow* (3rd ed.). O'Reilly Media.

2. McKinney, W. (2022). *Python for Data Analysis* (3rd ed.). O'Reilly Media.

3. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research, 12*, 2825–2830.

4. Hunter, J. D. (2007). Matplotlib: A 2D Graphics Environment. *Computing in Science & Engineering, 9*(3), 90–95.

5. Harris, C. R., et al. (2020). Array programming with NumPy. *Nature, 585*(7825), 357–362.

6. Kaggle. (n.d.). *Titanic - Machine Learning from Disaster*. https://www.kaggle.com/competitions/titanic

APPENDIX: PYTHON CODE

The following is the complete Python code used in the accompanying Jupyter Notebook to perform data acquisition, data cleaning, data visualisation, statistical analysis, and machine learning on the Titanic dataset.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Make plots look nice
sns.set(style="whitegrid")

# Load the dataset
df = pd.read_csv("titanic.csv")
print("Dataset loaded successfully")

# Dataset dimensions
print("Dataset dimensions (rows, columns):", df.shape)

# Column names
print("Column names:")
print(df.columns.tolist())

# First five observations
```

```python
df.head()

# Data types
df.dtypes

# Detect missing values
missing = df.isnull().sum()
missing_percent = (missing / len(df)) * 100
missing_table = pd.DataFrame({"Missing Count": missing, "Missing %":
missing_percent.round(2)})
missing_table[missing_table["Missing Count"] > 0]

# Handle missing values
df["Age"] = df["Age"].fillna(df["Age"].median())
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
df["Has_Cabin"] = df["Cabin"].notnull().astype(int)
df = df.drop(columns=["Cabin"])

# Confirm no more missing values in the columns we care about
df.isnull().sum()

# Detect duplicated observations
duplicate_count = df.duplicated().sum()
print("Number of duplicate rows:", duplicate_count)

# Remove duplicates if any exist
if duplicate_count > 0:
    df = df.drop_duplicates()
    print("Duplicates removed. New shape:", df.shape)
else:
    print("No duplicates found — no rows removed.")

plt.figure(figsize=(8,5))
plt.hist(df["Age"], bins=20, color="steelblue", edgecolor="black")
plt.title("Distribution of Passenger Ages")
plt.xlabel("Age (years)")
plt.ylabel("Number of Passengers")
plt.savefig("age_histogram.png", dpi=100, bbox_inches="tight")
plt.show()

plt.figure(figsize=(7,5))
df["Pclass"].value_counts().sort_index().plot(kind="bar",
color=["#4C72B0","#DD8452","#55A868"])
plt.title("Number of Passengers by Ticket Class")
plt.xlabel("Passenger Class (1 = 1st, 2 = 2nd, 3 = 3rd)")
plt.ylabel("Number of Passengers")
plt.xticks(rotation=0)
plt.savefig("class_bar.png", dpi=100, bbox_inches="tight")
plt.show()

plt.figure(figsize=(8,5))
sns.boxplot(x="Pclass", y="Age", data=df, hue="Pclass", palette="Set2", legend=False)
plt.title("Age Distribution Across Passenger Classes")
plt.xlabel("Passenger Class")
plt.ylabel("Age (years)")
plt.savefig("age_by_class_box.png", dpi=100, bbox_inches="tight")
plt.show()
```

```
plt.figure(figsize=(8,5))
plt.scatter(df["Age"], df["Fare"], alpha=0.5, color="darkorange")
plt.title("Age versus Fare Paid")
plt.xlabel("Age (years)")
plt.ylabel("Fare (£)")
plt.savefig("age_fare_scatter.png", dpi=100, bbox_inches="tight")
plt.show()

numeric_df = df.select_dtypes(include=[np.number])
corr = numeric_df.corr()

plt.figure(figsize=(9,7))
sns.heatmap(corr, annot=True, fmt=".2f", cmap="coolwarm", center=0)
plt.title("Correlation Heatmap of Numerical Variables")
plt.savefig("correlation_heatmap.png", dpi=100, bbox_inches="tight")
plt.show()

selected_cols = ["Survived", "Pclass", "Age", "Fare"]
sns.pairplot(df[selected_cols], hue="Survived", palette="husl", diag_kind="hist")
plt.suptitle("Pairplot of Survived, Pclass, Age, and Fare", y=1.02)
plt.savefig("pairplot.png", dpi=100, bbox_inches="tight")
plt.show()

df.describe()

print("Survival counts:")
print(df["Survived"].value_counts())
print("\nSex counts:")
print(df["Sex"].value_counts())
print("\nEmbarked counts:")
print(df["Embarked"].value_counts())
print("\nPclass counts:")
print(df["Pclass"].value_counts())

corr_with_survived = corr["Survived"].sort_values(ascending=False)
print(corr_with_survived)

strongest_positive = corr_with_survived.drop("Survived").idxmax()
strongest_positive_value = corr_with_survived.drop("Survived").max()
strongest_negative = corr_with_survived.drop("Survived").idxmin()
strongest_negative_value = corr_with_survived.drop("Survived").min()

print(f"Strongest positive correlation with Survived: {strongest_positive}
({strongest_positive_value:.2f})")
print(f"Strongest negative correlation with Survived: {strongest_negative}
({strongest_negative_value:.2f})")

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Encode categorical variables as numbers so the model can use them
le_sex = LabelEncoder()
df["Sex_encoded"] = le_sex.fit_transform(df["Sex"])      # male=1, female=0 (alphabetical)

le_embarked = LabelEncoder()
df["Embarked_encoded"] = le_embarked.fit_transform(df["Embarked"])
```

```
# Chosen predictor variables (features)
features = ["Pclass", "Sex_encoded", "Age", "SibSp", "Parch", "Fare", "Embarked_encoded",
"Has_Cabin"]
X = df[features]
y = df["Survived"]

print("Features used:", features)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
print("Training set size:", X_train.shape)
print("Testing set size:", X_test.shape)

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
print("Model trained successfully")

y_pred = model.predict(X_test)
print("Predictions on first 10 test samples:", y_pred[:10])

accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy: {accuracy:.4f} ({accuracy*100:.2f}%)")

cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(6,5))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
            xticklabels=["Did Not Survive","Survived"],
            yticklabels=["Did Not Survive","Survived"])
plt.title("Confusion Matrix")
plt.xlabel("Predicted Label")
plt.ylabel("Actual Label")
plt.savefig("confusion_matrix.png", dpi=100, bbox_inches="tight")
plt.show()

print("Classification Report:\n")
print(classification_report(y_test, y_pred, target_names=["Did Not Survive","Survived"]))
```